turned to glee, therefore, when I chanced upon an old stencilled bibliography that I had handed to the students at an Oxford lecture in 1970. Large as life, there was 'The genetical *theory* of social behaviour', a whole five years earlier than Wilson's publication. Wilson couldn't possibly have seen my 1970 bibliography. There was no doubt about it: Wilson and I had independently introduced the same mutant meme!

How could such a coincidence have happened? Once again, as in the case of 'Auld Lang Syne', a plausible explanation is not far to seek. R. A. Fisher's most famous book is called *The Genetical Theory of Natural Selection*. Such a household name has this title become in the world of evolutionary biologists, it is hard for us to hear its first two words without automatically adding the third. I suspect that both Wilson and I must have done just that. This is a happy conclusion for all concerned, since nobody minds admitting to being influenced by Fisher!

***p. 255* The computers in which memes live are human brains.**

It was obviously predictable that manufactured electronic computers, too, would eventually play host to self-replicating patterns of information—memes. Computers are increasingly tied together in intricate networks of shared information. Many of them are literally wired up together in electronic mail exchange. Others share information when their owners pass floppy discs around. It is a perfect milieu for self-replicating programs to flourish and spread. When I wrote the first edition of this book I was naïve enough to suppose that an undesirable computer meme would have to arise by a spontaneous error in the copying of a legitimate program, and I thought this an unlikely event. Alas, that was a time of innocence. Epidemics of 'viruses' and 'worms', deliberately released by malicious programmers, are now familiar hazards to computer-users all over the world. My own hard disc has to my knowledge been infected in two different virus epidemics during the past year, and that is a fairly typical experience among heavy computer-users. I shall not mention the names of particular viruses for fear of giving any nasty little satisfaction to their nasty little perpetrators. I say 'nasty', because their behaviour seems to me morally indistinguishable from that of a technician in a microbiology laboratory, who deliberately infects the drinking water and seeds epidemics in order to snigger at people getting ill. I say 'little', because these people are mentally little. There is nothing clever about designing a computer virus. Any half-way competent programmer could do it, and half-way competent programmers are two-a-penny in the modern world. I'm one myself. I shan't even bother to explain how computer viruses work. It's only too obvious.

What is less easy is to know how to combat them. Unfortunately some very expert programmers have had to waste their valuable time writing virus-detector programs, immunization programs, and so on (the analogy with medical vaccination, by the way, is astonishingly close, even down to the injection of a 'weakened strain' of the virus). The danger is that an arms race will develop, with each advance in virus-prevention being matched by counter-advances in new virus programs. So far, most anti-virus programs are written by altruists and supplied free of charge as a service. But I foresee the growth of a whole new profession—splitting into lucrative specialisms just like any other profession—of 'software doctors', on call with black bags full of diagnostic and curative floppy discs. I use the name 'doctors', but real doctors are solving natural problems that are not deliberately engineered by human malice. My software doctors, on the other hand, will be, like lawyers, solving man-made problems that should never have existed in the first place. In so far as virus-makers have any discernible motive, they presumably feel vaguely anarchistic. I appeal to them: do you really want to pave the way for a new fat-cat profession? If not, stop playing at silly memes, and put your modest programming talents to better use.

***p. 257* Blind faith can justify anything.**